

Covert authentication for securing MIMO communications

by

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The student author, whose presentation of the scholarship herein was approved by the program of study committee, is solely responsible for the content of this thesis. The Graduate College will ensure this thesis is globally accessible and will not permit alterations after a degree is conferred.

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DEDICATION

I would like to dedicate this thesis to my wife Soo-Young Kwak without whose support I would not have been able to complete this work.

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NOMENCLATURE

$\mathbf{x}$	Vector
$\mathbf{X}$	Matrix
$\mathbf{X}$	Determinant of $\mathbf{X}$
$[\mathbf{X}]_{i,j}$	Element in i-th row and j-th column of $\mathbf{X}$
$\mathbf{X}^H$	Hermitian transpose of $\mathbf{X}$
$\hat{x}$	Estimated value of x
$ x $	Absolute value of the scalar x
$\ \mathbf{x}\ $	L-2 norm of $\mathbf{x}$
$\int$	Integral operator
$\sum$	Summation operator
$\in$	Member of
$(\cdot, \cdot)$	Range of values
$\mathbb{E}[\cdot]$	Expectation operator
$\chi^2(k)$	Chi-squared distribution with k degree of freedom
$\mathcal{CN}(\mu, \sigma^2)$	Complex circularly symmetric Gaussian random variable with mean μ and variance σ^2
$Pr(\cdot)$	Probability
$f_X(x)$	Probability density function of X

$F_X(x)$ Cumulative density function of X

$Tr[\cdot]$ Trace operator

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ABSTRACT

This thesis aims to propose a covert authentication scheme that remains obfuscated from potential eavesdroppers by hiding the existence of the authentication tag. The attackers' endeavor to impersonate a legitimate transmitter can be countered by generating the invisible tag. The essential concept we suggest is to superimpose the low-power tag onto the message such that we can effectively avoid the attention of malicious users. We derive the total detection error probability and demonstrate that it can converge to 1, implying that the tag is not detectable by the malicious users, as the number of transmit antennas increases. Also, the covert tag and authenticated message throughput are obtained and analyzed to see the influence of the number of antennas.

COVERT AUTHENTICATION FOR SECURING MIMO COMMUNICATIONS

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1 Introduction

Among the numerous methods employed to bolster security in wireless communications, message authentication plays a fundamental role in ensuring secure communications. It addresses two aspects in communications that are message integrity and source authenticity. Traditionally, appending a cryptographic authentication tag to the message is one of the approaches [Menezes et al. \(2018\)](#). However, due to the open medium of wireless communications, there always exists a risk of malicious actors who attempt to extract the secret key from the message and tag pair by using cryptanalysis algorithms or brute-force attacks with quantum computing.

To mitigate the aforementioned vulnerability, it becomes imperative to employ strategies aimed to obscure the authentication tag from endeavor for the detection by malicious users. In this regard, the superimposition of low-power authentication tags, known as fingerprint embedding, in single-input single-output (SISO) and multi-input multi-output (MIMO) channels has been proposed in [Paul et al. \(2008\)](#) and [Paul and Sadler \(2011\)](#), respectively. [Verma et al. \(2015\)](#) employs a software-defined radio to investigate the probability of authentication when the low-power authentication tag is embedded. In [Chalise and Sadler \(2019\)](#), the mean-square error (MSE) of the authentication tags is derived based on the data and tag precoder matrices. The MSE is minimized through optimization, rendering improved authentication probability at the receiver and enhanced key equivocation for potential eavesdroppers. [Zhang et al. \(2019\)](#) proposes

superimposing a lightweight authentication for Internet-of-Things (IoT) networks to ensure secure and reliable authentication within IoT networks, in which a multiple-antenna base station serves multiple single-antenna devices.

In this thesis, we propose a covert authentication technique in which the tag remains impervious to potential eavesdroppers, even with unlimited computing power by hiding the existence of the tag. By harnessing the invisible tag, the attacker's endeavor to impersonate the legitimate user becomes impossible. We effectively avoid the attention of malicious users toward the tag through the use of an invisible tag. Our work deviates from cryptographic protection of authentication tag and provides enhanced security. Fundamental concept involves superimposing the low-power authentication tag onto the message and exploiting multiple antennas in order to conceal the tag. The total detection error probability at a multiple-antenna eavesdropper is derived and we show it converges to 1, implying the covertness of the tag, as the number of transmit antennas increases. We also show the covert tag throughput and the authenticated message throughput; thereby, we analyze the impact of the number of transmit antennas on our proposed scheme.

In the rest of this chapter, Section 2 illustrates the system model. In Section 3, we derive the total detection error probability. Section 4 analyzes the decoding error probability of the authentication tag followed by the covert tag throughput. Section 5 derives the authenticated message throughput. Section 6 shows the numerical results, and Section 7 concludes this chapter.

2 System Model

We consider a MIMO wireless communication where there are a transmitter (Alice) and a receiver (Bob) with multiple antennas, N_A and N_B , respectively. The communication occurs with an eavesdropper (Eve) with N_E antennas who wants to overhear the messages exchanged between Alice and Bob. We assume that Alice has a message to transmit for L duration, i.e., the message is $\mathbf{c} = [c_1, \dots, c_L]$,

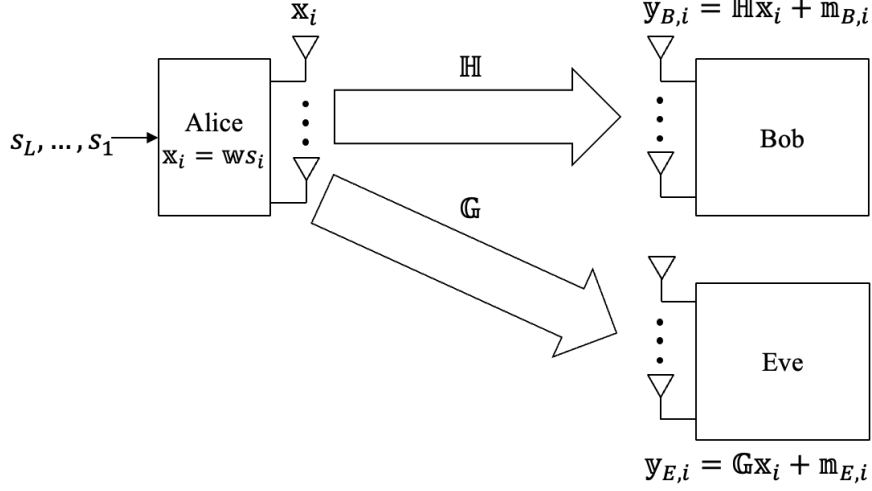


Figure 1 Proposed system model, in which a transmitter employs the MRT beamforming to transmit the messages while a legitimate receiver and an eavesdropper receive and combine the received signal at each antenna.

and Alice wants to ensure the integrity of the transmitted message by employing an authentication tag, derived from a cryptographic hash function, $h(\cdot)$, [Menezes et al. \(2018\)](#), such that the tag is $\mathbf{t} = h(\mathbf{c}, K)$ where K is the secret key exclusively shared between Alice and Bob. Usually, the length of output from $h(\cdot)$ is smaller than the message to authenticate. However, as adopted in [Paul and Sadler \(2011\)](#), we assume that Alice utilizes some modulation and coding scheme such that the authentication tag has the same length as the message; hence the tag is $\mathbf{t} = [t_1, \dots, t_L]$. By superimposing a low-power $\mathbf{t}$ onto $\mathbf{c}$, the tag can remain invisible from potential eavesdroppers while allowing Bob to authenticate the message. Hence, the transmitted signal, $\mathbf{s} \in \mathcal{C}^{1 \times L}$, is given by:

$$\mathbf{s} = \sqrt{\alpha} \mathbf{c} + \sqrt{1 - \alpha} \mathbf{t} \quad (1)$$

where $\alpha \in (0, 1)$ is the power allocation factor between $\mathbf{c}$ and $\mathbf{t}$. We assume that $\mathbb{E}[\mathbf{c}^H \mathbf{t}] = \mathbf{0}$, which is typically achieved by the hash function. Also, $\mathbf{c}$ and $\mathbf{t}$ independently generated codewords with $c_i, t_i \sim \mathcal{CN}(0, P)$ for $i = 1, \dots, L$ where P represents the power constraint. The transmission rate of $\mathbf{t}$ is fixed at R_t . We assume that the codebook for $\mathbf{s}$ is known to all users including the eavesdropper while $\mathbf{t}$'s codebook is shared only between Alice and Bob.

As for the channel model, we assume the channel between Alice and Bob is denoted by a channel matrix, $\mathbf{H} \in \mathcal{C}^{N_B \times N_A}$, and the channel between Alice and Eve is denoted by $\mathbf{G} \in \mathcal{C}^{N_E \times N_A}$. For brevity, we will call the channels Bob's channel and Eve's channel, respectively, hereafter. Therefore, the channel matrices can be represented with vectors that map the channels into each receive antenna, i.e.,

$$\mathbf{H} = [\mathbf{h}_1^T, \dots, \mathbf{h}_{N_B}^T]^T \quad (2)$$

$$\mathbf{G} = [\mathbf{g}_1^T, \dots, \mathbf{g}_{N_E}^T]^T \quad (3)$$

where $\mathbf{h}_i, \mathbf{g}_j \in \mathcal{C}^{1 \times N_A}$ for $i = 1, \dots, N_B$ and $j = 1, \dots, N_E$ are channel mappings each receive antenna, respectively. Since we assume Rayleigh fading channels, elements of the row vectors independently follow complex Gaussian distribution with zero mean and unit variance.

The received signal at Bob's receive antennas for L duration is given by:

$$\mathbf{y}_{B,i} = \mathbf{H}\mathbf{x}_i + \mathbf{n}_{B,i} \quad (4)$$

for $i = 1, \dots, L$, where $\mathbf{n}_{B,i}$ is the additive white Gaussian noise (AWGN) with zero mean and variance σ_B^2 . The transmitted signal from Alice's antennas for L duration is given by

$$\mathbf{x}_i = \mathbf{w}s_i \quad (5)$$

for $i = 1, \dots, L$ where $\mathbf{w} \in \mathcal{C}^{N_A \times 1}$ is the normalized transmit weighting vector that is defined as

$$\mathbf{w} = \frac{\mathbf{H}^H \mathbf{u}}{\|\mathbf{H}^H \mathbf{u}\|}. \quad (6)$$

$\mathbf{u} \in \mathcal{C}^{N_B \times 1}$ is the receive weighting vector at Bob. Bob can combine the signal received at each antenna by multiplying the receive weighting vector, $\mathbf{u}^H$. Hence, the combined signal is

$$r_{B,i} = \mathbf{u}^H \mathbf{y}_{B,i} \quad (7)$$

$$= \mathbf{u}^H \mathbf{H} \mathbf{x}_{B,i} + \mathbf{u}^H \mathbf{n}_{B,i} \quad (8)$$

$i = 1, \dots, L$. The SNR of (7) is maximized by selecting $\mathbf{u}$ as the eigenvector of $\mathbf{H}\mathbf{H}^H$ that corresponds to the largest eigenvalue of $\mathbf{H}\mathbf{H}^H$ [Dighe et al. \(2003\)](#). Hence, with the optimal

weighting vector, (8) becomes

$$r_{B,i} = \sqrt{\lambda_{(1)}} s_i + z_{B,i} \quad (9)$$

where $\lambda_{(1)}$ is the largest eigenvalue of $\mathbf{H}\mathbf{H}^H$ and $z_{B,i} \sim \mathcal{CN}(0, \sigma_B^2)$. The mean and variance of $z_{B,i}$ is preserved because $\|\mathbf{u}\|^2 = 1$. Therefore, we define the received signal for L observation as

$$\mathbf{r}_B = [r_{B,1}, \dots, r_{B,L}] \quad (10)$$

When Eve eavesdrops on the signal from Alice, the received signal at Eve's antennas is given by

$$\mathbf{y}_{E,i} = \mathbf{G}\mathbf{x}_i + \mathbf{n}_{E,i} \quad (11)$$

$i = 1, \dots, L$ where $\mathbf{n}_{E,i}$ is the AWGN noise with zero mean and variance σ_E^2 . We consider the worst case for legitimate users in which Eve is fully capable of estimating $\mathbf{g}_w := \mathbf{G}\mathbf{w}$ by using the blind channel estimation [Li \(2005\)](#). Hence, after Eve conducts the maximum ratio combining (MRC), the combined signal is given by

$$r_{E,i} = \mathbf{w}_E^H \mathbf{y}_{E,i} \quad (12)$$

$$= \|\mathbf{g}_w\| s_i + z_{E,i} \quad (13)$$

for $i = 1, \dots, L$, where $\mathbf{w}_E = \mathbf{g}_w / \|\mathbf{g}_w\|$ is the normalized receive weighting vector and $z_{E,i}$ is the AWGN with zero mean and variance σ_E^2 . As shown in [Tang et al. \(2016\)](#), the resulting received SNR is $\|\mathbf{g}_w\|\gamma$ where $\gamma = P/\sigma_E^2$ and $\|\mathbf{g}_w\| \sim \chi_2(2N_E)$. Therefore, we define the received signal at Eve for L observation as

$$\mathbf{r}_E = [r_{E,1}, \dots, r_{E,L}] \quad (14)$$

3 Total Detection Error Probability

In this section, we conduct the likelihood ratio test in order to detect the existence of the authentication tag. Next, we derive the total detection error probability defined as the sum of the false alarm and missed detection probability, i.e., $P_F + P_M$, which will serve as a metric to measure the covertness of the tag.

3.1 Likelihood Ratio Test (LRT)

In the likelihood ratio test (LRT) for detecting the tag [Axell et al. \(2012\)](#), the null hypothesis, $\mathcal{H}_0$, is "the tag does not exist" and the alternative hypothesis, $\mathcal{H}_1$, is "there exists the tag." The LRT is given by

$$\frac{Pr(\mathbf{r}_E \mid \|\mathbf{g}_w\|^2, \mathbf{c}, \mathcal{H}_1)}{Pr(\mathbf{r}_E \mid \|\mathbf{g}_w\|^2, \mathbf{c}, \mathcal{H}_0)} \underset{\mathcal{H}_1}{\overset{\mathcal{H}_0}{\leq}} \delta \quad (15)$$

where

$$Pr(\mathbf{r}_E \mid \|\mathbf{g}_w\|^2, \mathbf{c}, \mathcal{H}_1) = \left(\frac{1}{\pi(\sigma_E^2 + (1-\alpha)\|\mathbf{g}_w\|^2\gamma)} \right)^L e^{-\frac{\|\mathbf{r}_E - \|\mathbf{g}_w\|\sqrt{\alpha}\mathbf{c}\|^2}{\sigma_E^2 + (1-\alpha)\|\mathbf{g}_w\|^2\gamma}} \quad (16)$$

$$Pr(\mathbf{r}_E \mid \|\mathbf{g}_w\|^2, \mathbf{c}, \mathcal{H}_0) = \left(\frac{1}{\pi\sigma_E^2} \right)^L e^{-\frac{\|\mathbf{r}_E - \|\mathbf{g}_w\|\mathbf{c}\|^2}{\sigma_E^2}} \quad (17)$$

are the conditional probability density function (PDF) of Eve's observation, $\mathbf{r}_E$ under $\mathcal{H}_1$ and $\mathcal{H}_0$, respectively. δ is a detection threshold.

3.1.1 When Eve can decode $\mathbf{c}$

[Kim and Ta \(2021\)](#) shows that (15) can be reformulated by utilizing (16) and (17). As a result, the test statistic, Λ , when Eve successfully decodes $\mathbf{c}$, is given by

$$\Lambda = \frac{\frac{1}{L}\|\mathbf{r}_E - \|\mathbf{g}_w\|\mathbf{c}\|^2}{\sigma_E^2} - \frac{\frac{1}{L}\|\mathbf{r}_E - \|\mathbf{g}_w\|\sqrt{\alpha}\mathbf{c}\|^2}{\sigma_E^2 + (1-\alpha)\|\mathbf{g}_w\|^2P} \underset{\mathcal{H}_1}{\overset{\mathcal{H}_0}{\leq}} \delta_E \quad (18)$$

where δ_E is a detection threshold. The authors find out that Eve can make $P_F + P_M$ zero by properly choosing $\delta_E \in \left(\frac{(2\sqrt{\alpha}-2\alpha)\|\mathbf{g}_w\|^2P}{\sigma_E^2 + (1-\alpha)\|\mathbf{g}_w\|^2P}, \frac{(2-2\sqrt{\alpha})\|\mathbf{g}_w\|^2P}{\sigma_E^2} \right)$, which will make analysis of the total detection error probability simple as we will discuss.

3.1.2 When Eve cannot decode $\mathbf{c}$

Eve conducts a marginalized likelihood ratio test (MLRT) over the whole message codewords. Hence, the new test statistic, Λ' , when Eve fails to decode $\mathbf{c}$, is given by

$$\Lambda' = \frac{Pr(\mathbf{r}_E \mid \|\mathbf{g}_w\|^2, \mathcal{H}_1)}{Pr(\mathbf{r}_E \mid \|\mathbf{g}_w\|^2, \mathcal{H}_0)} \underset{\mathcal{H}_1}{\overset{\mathcal{H}_0}{\leq}} \delta'_E \quad (19)$$

where δ'_E is a new threshold when Eve cannot decode $\mathbf{c}$. The authors show that Eve always observes the same distributions in the MLRT regardless the tag is superimposed or not, i.e., $\Lambda' = 1$, rendering $P_F + P_M = 1$ in which $[P_F, P_M] = [0, 1]$ when Eve chooses $\delta'_E \leq \Lambda' = 1$ and $[P_F, P_M] = [1, 0]$ when Eve chooses $\delta'_E > \Lambda' = 1$.

3.2 Total Detection Error Probability

As per Shannon's channel coding theorem, a receiver is able to decode the received signal if and only if the channel capacity (or the achievable rate) is larger than or equal to the transmission rate. Hence, the total detection error probability, $P_F + P_M$, averaged over the events that Eve can or cannot decode $\mathbf{c}$ is given by

$$\begin{aligned} P_F + P_M = & Pr(\Lambda > \delta_E, I(\mathbf{c}; \mathbf{r}_E) \geq R_{(1)} \mid \mathcal{H}_0) + Pr(\Lambda' > \delta'_E, I(\mathbf{c}; \mathbf{r}_E) < R_{(1)} \mid \mathcal{H}_0) \\ & + Pr(\Lambda \leq \delta_E, I(\mathbf{c}; \mathbf{r}_E) \geq R_{(1)} \mid \mathcal{H}_1) + Pr(\Lambda' \leq \delta'_E, I(\mathbf{c}; \mathbf{r}_E) < R_{(1)} \mid \mathcal{H}_1) \end{aligned} \quad (20)$$

where

$$I(\mathbf{c}; \mathbf{r}_E) = \begin{cases} \log_2(1 + \|\mathbf{g}_w\|^2 \gamma), & \mathcal{H}_0 \\ \log_2(1 + \frac{\|\mathbf{g}_w\|^2 \alpha \gamma}{\|\mathbf{g}_w\|^2 (1 - \alpha) \gamma + 1}), & \mathcal{H}_1 \end{cases} \quad (21)$$

where $I(\mathbf{c}; \mathbf{r}_E)$ is the Eve's achievable rate of $\mathbf{c}$ given $\mathbf{r}_E$, and $R_{(1)}$ is the transmission rate of $\mathbf{c}$:

$$R_{(1)} = \log_2 \left(1 + \frac{\lambda_{(1)} \alpha \gamma}{\lambda_{(1)} (1 - \alpha) \gamma + 1} \right) \quad (22)$$

As discussed in Section 3.1, Eve can make the first and third terms in (20) zero by properly choosing δ_E . In order for Eve to minimize (20) in case she fails to decode the message (when $I(\mathbf{c}; \mathbf{r}_E) < R_{(1)}$) in (20), she will choose $\delta'_E < 1$ under $\mathcal{H}_0$ rather than $\delta'_E \geq 1$ under $\mathcal{H}_1$ because $Pr(I(\mathbf{c}; \mathbf{r}_E) < R_{(1)} \mid \mathcal{H}_0) < Pr(I(\mathbf{c}; \mathbf{r}_E) < R_{(1)} \mid \mathcal{H}_1)$. That is $I(\mathbf{c}; \mathbf{r}_E)$ given $\mathcal{H}_0$ is larger than $I(\mathbf{c}; \mathbf{r}_E)$ given $\mathcal{H}_1$; thus the probability that $R_{(1)}$ is larger than $I(\mathbf{c}; \mathbf{r}_E)$ given $\mathcal{H}_0$ is less likely than the other. Therefore, the total detection error probability minimized by Eve with the optimum δ_E

and δ'_E is given by

$$\xi = \min_{\delta'_E, \delta_E} P_F + P_M \quad (23)$$

$$= Pr(I(\mathbf{c}; \mathbf{r}_E) < R_{(1)} \mid \mathcal{H}_0) \quad (24)$$

$$= Pr\left(\log_2(1 + \|\mathbf{g}_w\|^2 \gamma) < \log_2\left(1 + \frac{\lambda_{(1)} \alpha \gamma}{\lambda_{(1)}(1 - \alpha)\gamma + 1}\right)\right) \quad (25)$$

$$= Pr\left(\|\mathbf{g}_w\|^2 < \frac{\lambda_{(1)} \alpha}{\lambda_{(1)}(1 - \alpha)\gamma + 1}\right) \quad (26)$$

When we isolate $\lambda_{(1)}$, we only consider $\|\mathbf{g}_w\|^2 \leq \frac{\alpha}{(1-\alpha)\gamma}$ because, for $\|\mathbf{g}_w\|^2 > \frac{\alpha}{(1-\alpha)\gamma}$, (26)

becomes $Pr(\lambda_{(1)} < \frac{\|\mathbf{g}_w\|^2}{\alpha - \|\mathbf{g}_w\|^2(1-\alpha)\gamma}) = 0$ since always $\lambda_{(1)} \geq 0$. Hence, (26) can be rewritten as,

ξ

$$= Pr\left(\lambda_{(1)} > \frac{\|\mathbf{g}_w\|^2}{\alpha - \|\mathbf{g}_w\|^2(1-\alpha)\gamma} \mid \|\mathbf{g}_w\|^2 < \frac{\alpha}{(1-\alpha)\gamma}\right) Pr\left(\|\mathbf{g}_w\|^2 < \frac{\alpha}{(1-\alpha)\gamma}\right) \quad (27)$$

$$= Pr\left(\|\mathbf{g}_w\|^2 \leq \frac{\alpha}{(1-\alpha)\gamma}\right) - Pr\left(\lambda_{(1)} < \frac{\|\mathbf{g}_w\|^2}{\alpha - \|\mathbf{g}_w\|^2(1-\alpha)\gamma}, \|\mathbf{g}_w\|^2 \leq \frac{\alpha}{(1-\alpha)\gamma}\right) \quad (28)$$

$$\begin{aligned} &= Pr\left(\|\mathbf{g}_w\|^2 \leq \frac{\alpha}{(1-\alpha)\gamma}\right) \\ &\quad - Pr\left(\lambda_1 \leq \frac{\|\mathbf{g}_w\|^2}{\alpha - \|\mathbf{g}_w\|^2(1-\alpha)\gamma}, \dots, \lambda_N \leq \frac{\|\mathbf{g}_w\|^2}{\alpha - \|\mathbf{g}_w\|^2(1-\alpha)\gamma}, \|\mathbf{g}_w\|^2 \leq \frac{\alpha}{(1-\alpha)\gamma}\right) \end{aligned} \quad (29)$$

$$= F_{\|\mathbf{g}_w\|^2}\left(\frac{\alpha}{(1-\alpha)\gamma}\right) - \int_0^{\frac{\alpha}{(1-\alpha)\gamma}} \prod_{i=1}^N F_{\lambda_i}\left(\frac{x}{\alpha - x(1-\alpha)\gamma}\right) f_{\|\mathbf{g}_w\|^2}(x) dx \quad (30)$$

where the third equality holds because the CDF of the largest eigenvalue can be represented with the product of CDFs of a generic unordered eigenvalue, i.e., $F_{\lambda_{(1)}}(x) = \prod_{i=1}^N F_{\lambda_i}(x)$ because $\lambda_{(1)} = \max\{\lambda_1, \dots, \lambda_N\}$, in which $F_{\lambda_i}(x)$ follows from Lemma 1. Since the CDF of $\|\mathbf{g}_w\|^2$ is given by

$$F_{\|\mathbf{g}_w\|^2}(x) = 1 - e^{-x} \sum_{k=0}^{N_E-1} \frac{x^k}{k!} \quad (31)$$

(30) becomes

$$\xi = 1 - e^{-\frac{\alpha}{(1-\alpha)\gamma}} \sum_{k=0}^{N_E-1} \frac{1}{k!} \left(\frac{\alpha}{(1-\alpha)\gamma}\right)^k - \int_0^{\frac{\alpha}{(1-\alpha)\gamma}} \prod_{i=1}^N F_{\lambda_i}\left(\frac{x}{\alpha - x(1-\alpha)\gamma}\right) f_{\|\mathbf{g}_w\|^2}(x) dx \quad (32)$$

Lemma 1 (PDF of a generic unordered eigenvalue of the uncorrelated central Wishart matrix [Zanella et al. \(2009\)](#)). *When λ_i is a generic unordered eigenvalue of the uncorrelated central Wishart matrix, the PDF of λ_i is given by*

$$f_{\lambda_i}(x) = K \sum_{n=1}^N \sum_{m=1}^N (-1)^{n+m} x^{n+m-2+X-N} e^{-x} |\mathbf{\Omega}_{\lambda_i}| \quad (33)$$

where $|\mathbf{\Omega}_{\lambda_i}|$ is the determinant of the matrix, $\mathbf{\Omega}_{\lambda_i}$, which entries are given by

$$[\mathbf{\Omega}_{\lambda_i}]_{i,j} = (\alpha_{ij}^{(n)(m)} + X - N)! N^{-\frac{1}{N-1}} \quad (34)$$

$$\alpha_{i,j}^{(n)(m)} = \begin{cases} i + j - 2, & \text{if } i < n \text{ and } j < m \\ i + j, & \text{if } i \geq n \text{ and } j \geq m \\ i + j - 1, & \text{otherwise} \end{cases} \quad (35)$$

and K is the normalizing factor that is given by

$$K = \frac{1}{\prod_{i=1}^X (N - i)! \prod_{j=1}^N (N - j)!} \quad (36)$$

The cumulative distribution function (CDF) from $f_{\lambda}(x)$ is given by

$$F_{\lambda_i}(x) = \int_0^x f_{\lambda_i}(x) dx \quad (37)$$

Details are introduced and derived in [Zanella et al. \(2009\)](#). $\square$

Remark: Alice and Bob can benefit from the diversity gain by utilizing the MRT due to X and N factors in Lemma 1. If Bob has a single antenna, i.e., $N_B = 1$, $\lambda_{(1)}$ in (26) will become $\|\mathbf{h}\|^2$, in which $\|\mathbf{h}\|^2 \sim \chi^2(2N_A)$. The special case with $N_B = N_E = 1$ corresponds to the results in [Kim and Ta \(2021\)](#).

4 Covert Tag Throughput

In this section, we first derive the decoding outage probability of the authentication tag and analyze the covert tag throughput for a given tag transmission rate, R_t , based on the decoding outage probability.

4.1 Decoding Error Probability of Authentication Tag

From the previous work [Kim and Ta \(2021\)](#), if Bob fails to decode and subtract $\mathbf{c}$ from the received signal, it will be only noise that interrupts when Bob tries to decode $\mathbf{t}$. Otherwise, $\mathbf{c}$ will also behave as noise. Therefore, the achievable covert tag throughput of Bob's channel when he fails to decode and subtract $\mathbf{c}$ is $I(\mathbf{t}; \mathbf{r}_B)_0 = \log_2 \left(1 + \frac{\lambda_{(1)}(1-\alpha)\gamma}{\lambda_{(1)}\alpha\gamma+1} \right)$. On the other hand, when Bob succeeds in decoding and removing $\mathbf{c}$ from the received signal, only the noise will affect when decoding $\mathbf{t}$; hence, the achievable covert tag throughput when Bob successfully decodes $\mathbf{c}$ is $I(\mathbf{t}; \mathbf{r}_B)_1 = \log_2 (1 + \lambda_{(1)}(1 - \alpha)\gamma)$. The achievable message throughput is $I(\mathbf{c}; \mathbf{r}_B) = \log_2 \left(\frac{\lambda_{(1)}\alpha\gamma}{\lambda_{(1)}(1-\alpha)\gamma+1} \right)$. As a result, for a given tag transmission rate, R_t , the decoding outage probability of $\mathbf{t}$ averaged over events when Bob can and cannot decode $\mathbf{c}$ is given by

$$P_{o,t} = Pr(I(\mathbf{t}; \mathbf{r}_B)_0 < R_t, I(\mathbf{c}; \mathbf{r}_B) < R_{(1)}) + Pr(I(\mathbf{t}; \mathbf{r}_B)_1 < R_t, I(\mathbf{c}; \mathbf{r}_B) \geq R_{(1)}) \quad (38)$$

In our model, $I(\mathbf{c}; \mathbf{r}_B) \geq R_{(1)}$ in the second term of (38) always holds because $I(\mathbf{c}; \mathbf{r}_B) = R_{(1)}$.

Thus, (38) can be approximated as

$$P_{o,t} \simeq Pr(I(\mathbf{t}; \mathbf{r}_B)_1 < R_t) \quad (39)$$

$$= Pr(\log_2 (1 + \lambda_{(1)}(1 - \alpha)\gamma) < R_t) \quad (40)$$

$$= Pr\left(0 \leq \lambda_{(1)} < \frac{2^{R_t} - 1}{\gamma(1 - \alpha)}\right) \quad (41)$$

$$= \prod_{i=1}^N F_{\lambda_i} \left(\frac{2^{R_t} - 1}{\gamma(1 - \alpha)} \right) \quad (42)$$

4.2 Covert Tag Throughput

Following the result of the decoding outage probability of the authentication tag, the covert tag throughput, η_t , for a given R_t subject to the constraint that $\xi \geq 1 - \epsilon$ is given by

$$\eta_t = (1 - P_{o,t})R_t \quad (43)$$

$$= \left(1 - \prod_{i=1}^N F_{\lambda_i} \left(\frac{2^{R_t} - 1}{\gamma(1 - \alpha)} \right) \right) R_t \quad (44)$$

$$\text{s.t. } \xi \geq 1 - \epsilon$$

where ϵ is an arbitrary positive small constant, and $(1 - P_{o,t})$ explains that the received authentication tag is not in an outage.

5 Authenticated Message Throughput

In this section, we analyze the average message throughput given that the authentication tag is superimposed onto it. From (22), the average message throughput is

$$\eta_s = \int_0^\infty \log_2 \left(1 + \frac{x\alpha\gamma}{x(1-\alpha)\gamma + 1} \right) f_{\lambda_{(1)}}(x) dx \quad (45)$$

where $f_{\lambda_{(1)}}(x)$ follows from Lemma 2. For the average received message bits in (45), $1 - P_{o,t}$ are authenticated correctly. Hence, the authenticated message throughput is given by

$$W = \eta_s(1 - P_{o,t}) \quad (46)$$

Lemma 2 (PDF of the largest eigenvalue of uncorrelated central Wishart matrix [Zanella et al. \(2009\)](#)). *When $\lambda_{(1)}$ is the largest eigenvalue of the uncorrelated central Wishart matrix, the PDF of $\lambda_{(1)}$ is given by (33) where $\mathbf{\Omega}_{\lambda_i}$ is replaced with $\mathbf{\Omega}_{\lambda_{(1)}}$ which elements are defined as*

$$[\mathbf{\Omega}_{\lambda_{(1)}}]_{i,j} = \int_0^x u^{\alpha_{i,j}^{(n)(m)} + X - N} e^{-u}, du \quad (47)$$

The detailed derivation of Lemma 2 is provided in [Zanella et al. \(2009\)](#). $\square$

6 Numerical Results

In this section, we present the numerical results based on findings in the previous sections.

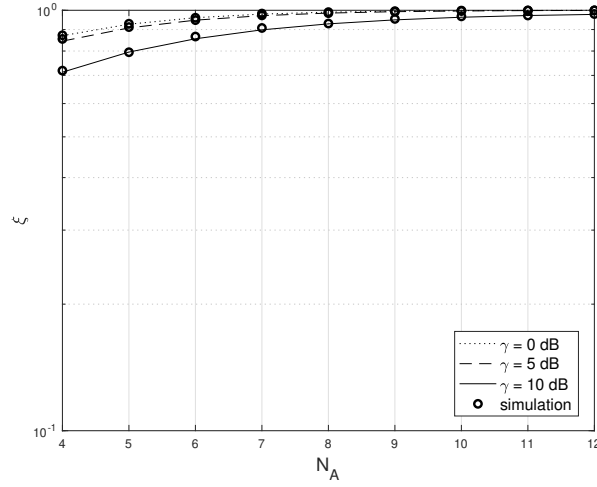


Figure 2 The total detection error probability of covert tag, ξ , versus the transmit antennas, N_A , for different transmit SNR, γ ; $\alpha = 0.99$ and $N_B = N_E = 4$.

6.1 Total Detection Error Probability

Fig. 2 shows the total detection error probability, ξ , with respect to the number of the transmit antennas, N_A , for different transmit SNR, γ , when $\alpha = 0.99$. We fix the number of receive antennas, N_B , and the number of the eavesdropper's antennas, N_E , as 4. It shows that ξ converges to 1 when the number of transmit antennas is sufficiently large. Also, one can observe that the total detection error probability decreases as γ gets large. It implies that, as the transmit power increases, the eavesdropper is more likely to detect the superimposed tag. However, if N_A is large enough, the detection likelihood will be low even for a higher SNR regime. The simulation corresponds to the analytic result.

Fig. 3 illustrates ξ with respect to the power allocation factor, α , for $N_A = 16, 8$, and 4, when $N_B = N_E = 4$ and $\gamma = 0$ dB. It shows that the more power is allocated to the tag, i.e., α decreases, the more likely the tag is to be detected by the eavesdropper. However, the transmitter can keep the tag detection likelihood effectively low by increasing the number of transmit antennas in spite of a higher SNR regime. Hence, we can conclude that, even though there exists a trade-off between the power allocation factor and the total detection error probability, the

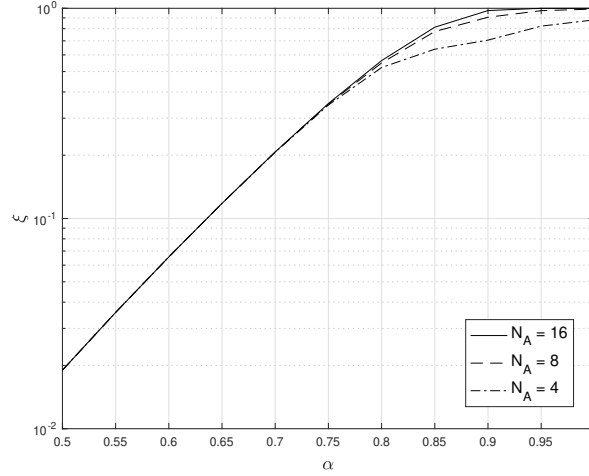


Figure 3 The total detection error probability, ξ , versus the transmit SNR, γ , for the different number of antennas, N ; $\alpha = 0.999$.

transmitter can offset the loss of the covertness by having a number of transmit antennas, ensuring the undetectability of the tag.

6.2 Decoding Outage Probability of Tag

Fig. 4 depicts the decoding outage probability of the authentication tag, $P_{o,t}$, with respect to the tag transmission rate, R_t , for $N_A = 4$ and 16. We fix parameters as $\alpha = 0.99$, $N_B = N_E = 4$, and $\gamma = 10$ dB. One can observe that as N_A increases, the decoding outage probability also increases. However, having more transmit antennas reduces the decoding outage probability of the tag. It indicates that increasing N_A enables the transmitter to send the tag more reliably. The simulation result corresponds to the analytic result.

6.3 Covert Tag Throughput

Fig. 5 depicts the covert tag throughput, η_t , with respect to the tag transmission rate, R_t , for $\alpha = 0.999, 0.99$, and 0.9. We fix parameters as $\gamma = 5$ dB, $N_A = 16$, $N_B = N_E = 4$, and the covertness constraint, denoted as ϵ , is 0.01. It reveals the existence of an optimum tag transmission rate that maximizes the covert tag throughput. Moreover, one can find out that the

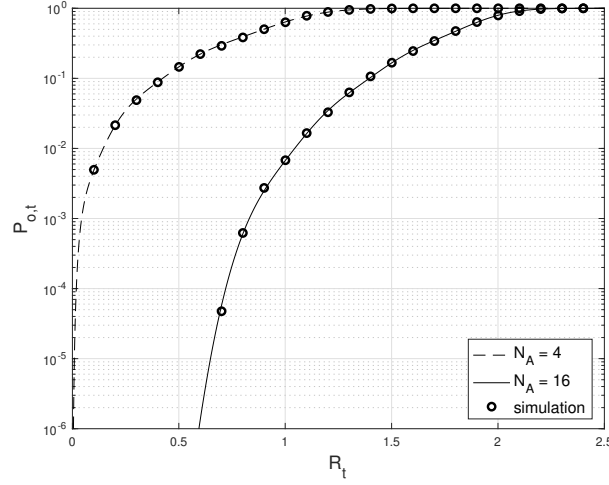


Figure 4 The decoding outage probability of the covert tag, $P_{o,t}$, versus the tag transmission rate, R_t , for $N_A = 4$ and 16; $\alpha = 0.99$ and $\gamma = 10$ dB.

more power is allocated to the authentication tag, the better covert tag throughput the system can provide. Hence, one can conclude that it is possible to obtain a higher covert tag throughput and a low probability of tag detection simultaneously by managing the power allocation factor.

6.4 Authenticated Message Throughput

Figure 6 presents the authenticated message throughput, W , with respect to the detection error probability constraint, ϵ , that is varied by the power allocation factor, α , over the range $[0, 1]$. The analysis is conducted for different numbers of transmit antennas, $N_A = 4, 8$, and 16, while keeping $N_B = N_E = 4$ and $R_t = 1$ fixed. The authenticated message throughput, W , represents the amount of data that can be transmitted reliably and authenticated by the legitimate receiver. A smaller value of ϵ indicates a higher level of covertness of the tag. One can observe that the authenticated throughput decreases as we impose a more strict covertness constraint on the tag. This decrement occurs because the tight tag covertness demands a reduction in the power that is allocated to the tag; thus, the probability of decoding the tag decreases. As a result, the probability of successfully authenticating the received message also decreases. However, it is possible for the transmitter to decrease the decoding outage probability

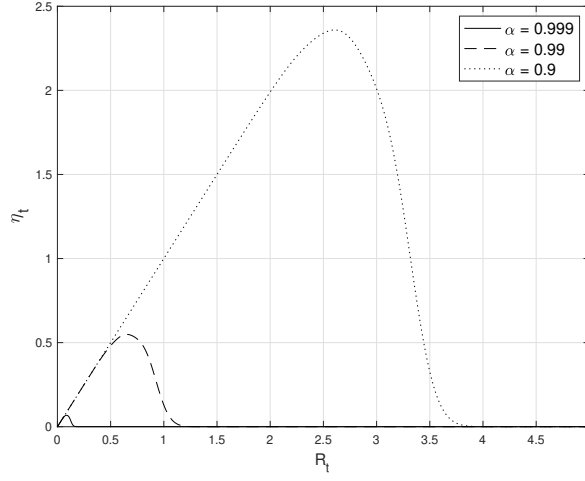


Figure 5 The covert tag throughput, η_t , versus the tag transmission rate, R_t , for different tag power allocation factor, α ; $N_A = 16$, $N_B = N_E = 4$, $\epsilon = 0.01$, and $\gamma = 5$ dB.

of the tag such that the authenticated message throughput is improved, by employing more antennas at the transmitter. The employment of a higher number of transmit antennas enables the system to decode the embedded tag more accurately, leading to improvement of the efficiency of message authentication and transmission.

7 Conclusion

In this thesis, we have explored the concept of covert authentication for improving the security of MIMO communications under Rayleigh fading potential eavesdropping attacks by effectively concealing the authentication tag. We derived the total detection error probability at an eavesdropper by harnessing the likelihood ratio test employed by malicious users to detect the existence of the tag. Based on the finding, we derived two important metrics, which are the covert tag throughput and the authenticated message throughput.

Throughout our study, we examined the impact of SNR and the extent of power allocation to the tag on the proposed scheme. Our findings reveal that the authentication tag can remain invisible from potential eavesdroppers who attempt to obtain the secret key information for the

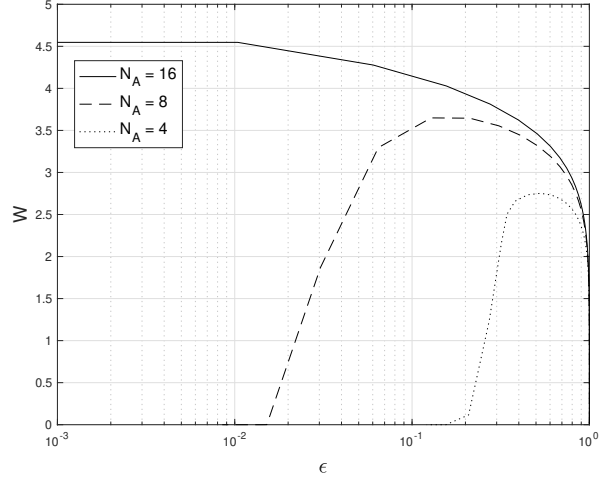


Figure 6 The authenticated message throughput, W , versus the tag covertness constraint, ϵ ; $R_t = 1$, $N_A = 16$, $N_B = N_E = 4$, $\alpha = 0.99$, and $\gamma = 5$ dB.

tag, while the message can be reliably transmitted and authenticated by the intended receiver if the number of transmit antennas is sufficiently large. These findings provide valuable insights into the reliability of the proposed covert authentication scheme.

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